

6.6(a)  $U_{I1}, U_{I2}$  接地时:

$$U_{O3} = \frac{R_f}{R_3} U_{I3} = 5 U_{I3}$$

$$\text{当 } U_{I3} \text{ 接地时, } U_{O1} = -\frac{R_f}{R_1} U_{O1} - \frac{R_f}{R_2} U_{O2} = -2U_{O1} - 2U_{O2}$$

$$\therefore U_O = -2U_{I1} - 2U_{I2} + 5U_{I3}$$

$$(b) U_{I1} \text{ 接地时, } U_{O1} = \frac{R_f}{R_2} U_{I2} + \frac{R_f}{R_3} U_{I3} = 10U_{I2} + U_{I3}$$

$$\text{当 } U_{I2}, U_{I3} \text{ 接地时, } U_{O2} = -\frac{R_f}{R_1} U_{I1} = -10U_{I1}$$

$$\therefore U_O = -10U_{I1} + 10U_{I2} + U_{I3}$$

$$(c) U_O = \frac{R_f}{R_1} (U_{I2} - U_{I1}) = 8(U_{I2} - U_{I1})$$

$$(d) U_O = -\frac{R_f}{R_1} U_{I1} - \frac{R_f}{R_2} U_{I2} + \frac{R_f}{R_3} U_{I3} + \frac{R_f}{R_4} U_{I4}$$

$$= -20U_{I1} - 20U_{I2} + 40U_{I3} + U_{I4}$$

$$6.8. \text{ 由“虚断”, } I_L = I_R = \frac{(V_{CC} - U_Z)}{R} = \frac{U_-}{R_2}$$

$$\text{由“虚短”, } U_+ = U_- = 6V$$

$$\therefore I_L = 0.6 \text{ mA}$$

$$\therefore R_2 = \frac{U_-}{I_L} = \frac{U_Z}{I_L}$$

$\therefore$  若  $I_L$  在  $1 \sim 10 \text{ mA}$  变化, 则  $R_2$  应改为用一个  $0.6 \text{ k}\Omega$  的电阻与一个  $5.4 \text{ k}\Omega$  的电位器串联。

$$6.9 (1). U_{2+} = U_{2-} = \frac{R_f}{R} (U_{I2} - U_{I1}) = 10(U_{I2} - U_{I1})$$

$$\therefore U_O = \frac{R_w}{R_1} U_{2+} = 10 \cdot \frac{R_w}{R_1} (U_{I2} - U_{I1})$$

12). 由(1) 知此时  $R_1 = R_w$

$$\therefore U_0 = 100 \text{ mV}$$

$$(3) \therefore U_0 = 10 \frac{R_w}{R_1} (U_{I2} - U_{I1}) = \frac{100 \text{ k}\Omega}{10 \text{ k}\Omega - R_2} (U_{I2} - U_{I1})$$

$$|U_0| = \frac{100}{10 - R_2} |U_{I2} - U_{I1}| \leq 14, |U_{I2} - U_{I1}|_{\max} = 20 \text{ mV}$$

$$\therefore R_2 \leq 9.86 \text{ k}\Omega, (R_2)_{\max} = 9.86 \text{ k}\Omega$$

$$6.10. (a) U_{I1} \text{ 接地}, U_{O2} = - \frac{R_3 + R_4}{R_2} (1 + \frac{R_3 // R_4}{R_5}) U_{I2};$$

$$U_{I2} \text{ 接地}, U_{O1} = - \frac{R_3 + R_4}{R_1} (1 + \frac{R_3 // R_4}{R_5}) U_{I1}$$

$$\therefore U_0 = - (R_3 + R_4) (1 + \frac{R_3 // R_4}{R_5}) (\frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2})$$

(b)  $U_{O1} = (1 + \frac{R_3}{R_1}) U_I$ , 假加在  $A_1$  同相输入端的信号为  $U_{I1}$ , 加在  $A_2$  同相输入端的信号为  $U_{I2}$ ,

$$U_I = U_{I1} - U_{I2}$$

$$\text{则 } U_{O1} = (1 + \frac{R_3}{R_1}) U_{I1}$$

$$U_0 = \frac{R_5}{R_3} U_{I2} - \frac{R_5}{R_4} U_{O1}$$

$$= (1 + \frac{R_5}{R_4}) U_{I2} - \frac{R_5}{R_4} (1 + \frac{R_3}{R_1}) U_{I1}$$

$$= \frac{R_5}{R_4} (1 + \frac{R_5}{R_4}) (U_{I2} - U_{I1})$$

$$= - (1 + \frac{R_5}{R_4}) U_I$$

(c)  $A_1, A_2, A_3$  的输出信号分别为  $U_{I1}, U_{I2}, U_{I3}$

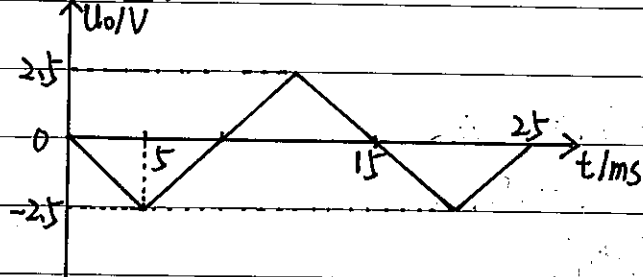
$\therefore$  对  $A_4$ ,

$$\begin{cases} \frac{U_{I1} - U_+}{R_1} + \frac{U_{I2} - U_+}{R_{a1}} + \frac{U_{I3} - U_+}{R_1} = \frac{U_+}{R_2} \\ \frac{U_0 - U_-}{R_4} = \frac{U_-}{R_3} \\ U_+ = U_- \end{cases}$$

$$\therefore U_0 = 10(U_{I1} + U_{I2} + U_{I3})$$

$$6.11 \quad U_0 = -\frac{1}{RC} \int u_i dt = -100 \int u_i dt$$

$\therefore$  如图



6.13 (a) 由“虚短”和“虚断”

$$\frac{U_I - 0}{R_1} = \frac{0 - (U_c + U_0)}{R_2} = i_c = i_R$$

$$\begin{aligned} U_0 &= -U_c - U_{0I} = -\frac{1}{C} \int i_c dt - U_I \\ &= -\frac{1}{RC} \int U_I dt - U_I = -U_I - 1000 \int U_I dt \end{aligned}$$

(b) 由“虚短”和“虚断”

$$\begin{cases} U_{c1} = U_{I1} - U_- = U_{I1} = \frac{1}{C_1} \int i_{c1} dt \\ i_{c1} = i_R = i_{c2} = \frac{0 - (U_{c2} + U_0)}{R} \end{cases}$$

$$\begin{aligned} \therefore U_0 &= -R i_{c1} - U_{c2} = -R i_{c1} - \frac{1}{C_2} \int i_{c2} dt \\ &= -R i_{c1} - \frac{1}{C_2} \int i_{c1} dt \\ &= -RC_1 \frac{dU_{I1}}{dt} - \frac{C_1}{RC_2} U_I \\ &= -10^{-3} \frac{dU_I}{dt} - 2U_I \end{aligned}$$

(c). 由“虚短”和“虚短断”

$$\begin{cases} \frac{U_- - U_+}{R} = i_{c1} = C \frac{dU_{c1}}{dt} = C \frac{d(U_0 - U_+)}{dt} \\ \frac{U_I - U_+}{R} = i_{c2} = C \frac{dU_{c2}}{dt} = C \frac{dU_+}{dt} \end{cases}$$

$$\therefore U_0 = \frac{1}{RC} \int U_I dt = 10^3 \int U_I dt$$

$$\begin{aligned} \text{(d)} \quad U_0 &= -\frac{1}{C} \int \left( \frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right) dt \\ &= -100 \int (U_{I1} + 0.5 U_{I2}) dt \end{aligned}$$

6.14. (1) 对  $A_1$ , 由电阻对称性可知

$$U_{O1} = \frac{R_4}{R_1} (U_0 - U_{I1}) = U_0 - U_{I1}$$

对  $A_2$ , 由“虚短”和“虚断”

$$\frac{U_{O1} - U_+}{R} = i_c = C \frac{du_c}{dt} = C \frac{du_-}{dt}$$

$$\because U_+ = U_- = U_0$$

$$\therefore \frac{-U_{I1}}{R} = C \frac{dU_0}{dt}$$

$$\therefore U_0 = -\frac{1}{RC} \int U_I dt = -10 \int U_I dt$$

$$(2) \because U_0 = -10 \int U_I dt = 10t + U_0(0) = 10t$$

$$\therefore U_0 = 6V \text{ 所需时间 } t = 0.6s$$

6.15 对  $A_2$ , 输出端电压为

$$U_{O2} = \left( 1 + \frac{R_2}{R_3} \right) U_0 = 2U_0$$

$$\text{对 } A_1, \quad U_{O2} = -\frac{1}{RC} \int U_I dt = -2 \int U_I dt$$

$$\therefore U_0 = -\int U_I dt$$

6.16 (1) 由“虚短”, “虚断”

$$U_B = U_{I1} = 4V, U_C = U_{I2} = 1V, i_A = \frac{U_B - U_C}{R}$$

$$U_A = U_B + R \cdot i_A = 7V,$$

$$U_D = U_C - R \cdot i_A = -2V.$$

对  $A_3, A_4$ ,

$$\begin{cases} \frac{U_A - U_0}{R} = \frac{U_0 - U_3}{R} & U_{03} = U_3 - = U_{3+} = 0V. \\ \frac{U_0 - U_4}{R} = \frac{U_4 - U_0}{R} \\ U_{4+} = U_D \end{cases}$$

$$\therefore U_0 = 2U_D = -4V.$$

(2) 当开关断开时,

$$\text{对 } A_3, \quad U_{03} = -\frac{1}{RC} \int U_A dt = -\frac{20}{0.05} \int U_A dt.$$

对  $A_4$ ,  $A_4$  实现减法运算.

$$\begin{aligned} U_0 &= -\frac{R}{R} + 2U_D - U_{03} = -4 + \frac{20}{0.05} \int U_A dt \\ &= -4 + \frac{140}{0.05} t + U_0(0) = 0.28t \end{aligned}$$

$\therefore$  当  $U_0 = 0$  时  $t \approx 28.6ms$ .

6.18 (a) 对  $A$ ,  $U_0' = -R_3 \left( \frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right) = 0.1 U_{I3} U_0.$

$$\therefore U_0 = \frac{-10R_3}{U_{I3}} \left( \frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right).$$

(b)  $A$  实现了反相求和运算, 三个输入电压分别为  $U_I, kU_I^2, k^2U_I^3$

$$\therefore U_0 = -\frac{R_4}{R_1} U_I - \frac{R_4}{R_2} k U_I^2 - \frac{R_4}{R_3} k^2 U_I^3$$